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Module 2.1 The National and International Economy

1. Measuring National Income — GDP

Gross Domestic Product (GDP) is the total value of all goods and services produced within a country's borders in a given time period (usually one year). It is the most commonly used measure of the size and performance of an economy.

1.1 Three Methods of Measuring GDP

Method	What it Measures	Explanation
Output (Production)	Value added at each stage of production	Sum of value added by all industries. Avoids double-counting by only counting the value added at each stage, not gross output.
Income	Total incomes earned by factors of production	Sum of wages, rent, interest, and profit earned in the economy.
Expenditure	Total spending on final goods and services	C + I + G + (X − M). Consumer spending + Investment + Government spending + Net exports.

All three methods should give the same figure (the circular flow of income means output = income = expenditure).

1.2 Nominal vs Real GDP

Nominal GDP (money GDP) is measured at current prices — it includes the effect of inflation. **Real GDP** is adjusted for inflation (measured at constant/base-year prices). It shows the actual change in output.

$$\text{Real GDP} = \text{Nominal GDP adjusted for inflation (using a GDP deflator)}$$

Real GDP is the better measure for comparing output over time because it strips out the effect of price changes.

1.3 GDP per Capita

$$\text{GDP per capita} = \text{GDP} / \text{Population}$$

GDP per capita gives a better indication of average living standards than total GDP, as it accounts for population size. China has a larger total GDP than Switzerland, but Switzerland has a much higher GDP per capita.

1.4 Limitations of GDP as a Measure of Living Standards

- **Distribution of income:** GDP per capita is an average — it says nothing about inequality. A country could have high GDP per capita but extreme poverty alongside extreme wealth.
- **Informal/shadow economy:** Unrecorded economic activity (cash-in-hand work, subsistence farming, illegal trade) is excluded.
- **Non-marketed output:** Household production (childcare, DIY, cooking) is not counted.
- **Quality of life factors:** GDP ignores leisure time, environmental quality, health, crime, political freedom.
- **Negative externalities:** Pollution, congestion, and environmental degradation may increase GDP (clean-up costs counted as output) while reducing welfare.
- **Composition of output:** Military spending and disaster reconstruction boost GDP but may not improve living standards.
- **Purchasing Power Parity (PPP):** International comparisons should use PPP-adjusted GDP to account for different price levels across countries.

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1.5 Alternative Measures of Living Standards

Measure	What it Includes	Advantages
Human Development Index (HDI)	Life expectancy, education (mean/expected years of schooling), GNI per capita (PPP)	Broader than GDP; captures health and education
Inequality-adjusted HDI (IHDI)	HDI adjusted for inequality within each dimension	Accounts for distribution, not just averages
Genuine Progress Indicator (GPI)	GDP adjusted for inequality, environmental degradation, unpaid work, crime costs	Attempts to measure true welfare
Gini Coefficient	Measures income inequality (0 = perfect equality, 1 = perfect inequality)	Direct measure of distribution

2. Economic Growth

Economic growth is an increase in the real GDP of an economy over time. **Short-run growth** is an increase in actual output (using existing resources more fully). **Long-run growth** is an increase in the productive capacity of the economy (potential output — shown by an outward shift of the PPF or rightward shift of LRAS).

2.1 Causes of Economic Growth

Short-Run Growth (Actual Growth)

- Increase in aggregate demand (AD) — rising consumer confidence, lower interest rates, fiscal stimulus, increased exports
- Reduction in spare capacity — moving from inside the PPF toward the frontier

Long-Run Growth (Potential Growth)

- **Increase in quantity of factors:** Population growth, immigration (labour), investment in capital, discovery of natural resources
- **Increase in quality of factors:** Better education and training (human capital), technological innovation, improved infrastructure
- **Institutional improvements:** Rule of law, property rights, political stability, efficient financial systems

2.2 Benefits and Costs of Economic Growth

Benefits:

- Higher real incomes and living standards
- Lower unemployment (firms need more workers to produce more output)
- Increased tax revenue for governments (without raising tax rates)
- Reduced absolute poverty
- Greater business confidence and investment

Costs:

- Environmental degradation (pollution, resource depletion, climate change)
- Inequality may worsen — growth may benefit the wealthy disproportionately
- Inflation risk if AD grows faster than productive capacity
- Current account deterioration if growth sucks in imports
- Structural change may cause structural unemployment as old industries decline
- Diminishing marginal returns to consumption — beyond a certain income, extra GDP adds little to happiness

Case Study: UK Economic Growth Since 2008

The UK experienced its deepest recession since the 1930s during the 2008-09 financial crisis, with GDP falling by over 6%. Recovery was slow — real GDP did not return to its pre-crisis level until 2013. Growth averaged around 2% per year from 2013-2019. The COVID-19 pandemic then caused a 9.7% fall in GDP in 2020, the largest in over 300 years, followed by a strong rebound of 7.4% in 2021 as the economy reopened. However, growth has since remained sluggish at around 0.5-1%,

with the UK struggling with low productivity growth, high energy costs, and Brexit-related trade frictions.

2.3 The Output Gap

Output gap is the difference between actual GDP and potential GDP (the level of output the economy could produce at full capacity).
Positive output gap: Actual GDP > Potential GDP — the economy is overheating (inflationary pressure).
Negative output gap: Actual GDP < Potential GDP — spare capacity exists (unemployment, low inflation).

3. Inflation

Inflation is a sustained increase in the general price level over time. It is measured as the annual percentage change in a price index.
Deflation is a sustained decrease in the general price level.
Disinflation is a fall in the rate of inflation (prices still rising, but more slowly).

3.1 Measuring Inflation

- **Consumer Price Index (CPI):** The UK's main measure. Based on a basket of approximately 700 goods and services weighted by typical household spending patterns. The basket is updated annually to reflect changing consumption habits. The Bank of England targets CPI inflation of 2%.
- **Retail Price Index (RPI):** Older measure that includes mortgage interest payments and council tax. Tends to be higher than CPI. Still used for some purposes (e.g. rail fares, student loan interest).

Limitations of CPI

- Based on an "average" basket — does not reflect individual spending patterns
- Slow to capture new products and quality improvements

- Substitution bias — assumes fixed consumption patterns (consumers actually switch to cheaper alternatives)
- Housing costs are not fully captured (owner-occupier housing costs excluded from CPI)
- Different income groups experience different inflation rates

3.2 Causes of Inflation

Demand-Pull Inflation

Caused by excess aggregate demand — AD grows faster than aggregate supply. "Too much money chasing too few goods." The AD curve shifts right, pushing up the price level.

Causes: Consumer confidence boom, low interest rates, fiscal stimulus, asset price bubbles, rapid credit growth, currency depreciation (boosts export demand).

Cost-Push Inflation

Caused by increases in the costs of production, which reduce aggregate supply. The SRAS curve shifts left, pushing up prices while reducing output (stagflation).

Causes: Rising oil/commodity prices, wage increases above productivity, currency depreciation (raises import prices), higher indirect taxes, supply chain disruptions.

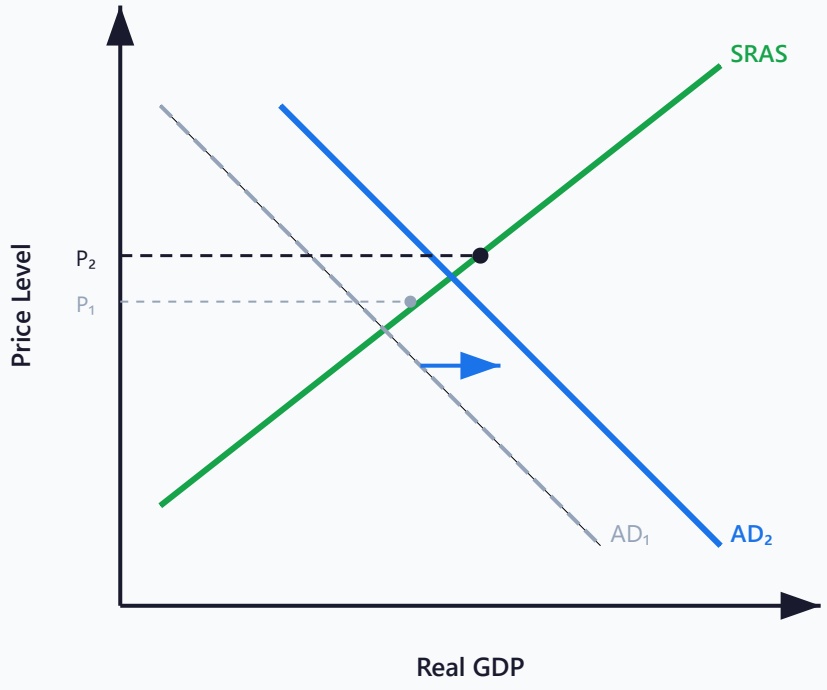

Figure 1: Demand-pull inflation — AD shifts right, raising the price level

3.3 Consequences of Inflation

- **Reduced purchasing power:** If wages do not rise as fast as prices, real incomes fall.
- **Menu costs:** Firms must frequently update prices (reprinting menus, reprogramming systems).
- **Shoe-leather costs:** People make more frequent trips to banks/ATMs to minimise cash holdings that lose value.
- **Redistribution:** Borrowers gain (repay in devalued money), savers lose (real value of savings falls), those on fixed incomes lose.
- **Uncertainty:** High/unpredictable inflation makes investment and planning difficult, reducing economic efficiency.
- **International competitiveness:** If UK inflation exceeds trading partners', exports become less competitive (worsening the current account).
- **Fiscal drag:** If tax thresholds are not inflation-adjusted, more people are pulled into higher tax bands.

Case Study: UK Inflation Surge 2021-2023

UK CPI inflation rose from 0.7% in early 2021 to a peak of 11.1% in October 2022 — the highest in 41 years. This was driven by a combination of: cost-push factors (global energy price surge after Russia's invasion of Ukraine, supply chain disruptions from COVID), demand-pull factors (post-lockdown consumer spending, fiscal stimulus), and imported inflation (sterling depreciation). The Bank of England responded by raising interest rates from 0.1% to 5.25% by August 2023. The episode illustrated how supply shocks and demand pressures can combine to create persistent inflation, and the difficulty of using monetary policy to address cost-push inflation without damaging growth.

4. Unemployment

Unemployment exists when people who are willing and able to work at the prevailing wage rate cannot find employment. The **unemployment rate** is the percentage of the labour force that is unemployed.

4.1 Measuring Unemployment

Measure	Method	Strengths	Weaknesses
Claimant Count	Number of people claiming unemployment-related benefits	Precise, timely, cheap to collect	Excludes those not eligible/claiming; affected by policy changes to benefit rules
Labour Force Survey (ILO)	Survey asking if people are without work, available for work, and actively seeking work	Internationally comparable, captures hidden unemployment	Sample survey (less precise), expensive, published with a lag

4.2 Types of Unemployment

Type	Cause	Example
Cyclical (demand-deficient)	Fall in aggregate demand during a recession — firms lay off workers as output falls	Mass redundancies during 2008-09 financial crisis
Structural	Mismatch between workers' skills/locations and available jobs — due to deindustrialisation, technology, or global shifts	Decline of coal mining, steel, shipbuilding in the UK
Frictional	Workers between jobs — normal and unavoidable in a dynamic economy	A graduate searching for their first job; someone who quit to find a better role
Seasonal	Regular fluctuations in demand for labour at certain times of year	Tourism workers, agricultural labourers, retail staff after Christmas

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Real wage (classical)	Wages set above the market-clearing level (e.g. by minimum wage or trade unions), creating excess supply of labour	If minimum wage is set significantly above equilibrium in a local labour market

4.3 Natural Rate of Unemployment (NRU)

Natural Rate of Unemployment (NRU) is the rate of unemployment that exists when the labour market is in equilibrium — consisting of frictional and structural unemployment only (no cyclical component). It is consistent with stable inflation. Also known as the **Non-Accelerating Inflation Rate of Unemployment (NAIRU)**.

4.4 Consequences of Unemployment

- **Individual:** Loss of income, skills atrophy (hysteresis), reduced self-esteem, mental health problems, social exclusion.
- **Government:** Higher benefit spending, lower tax revenue, increased public services demand (healthcare, policing).
- **Economy:** Lost output (economy inside PPF), reduced consumer spending, potential deflationary pressure.
- **Society:** Rising inequality, social unrest, higher crime rates, regional deprivation.

5. Balance of Payments

Balance of Payments (BoP) is a record of all monetary transactions between a country and the rest of the world over a period of time. It always balances overall (current account + capital account + financial account = 0).

5.1 Components

Account	What it Records	Key Items

Current Account	Trade in goods, trade in services, primary income, secondary income	Exports/imports of goods and services; investment income; transfers (aid, remittances)
Capital Account	Capital transfers and non-produced assets	Debt forgiveness, migrants' transfers, sale of patents
Financial Account	Investment flows — FDI, portfolio investment, other investment, reserve assets	Purchase of foreign businesses, shares, government bonds; central bank reserves

5.2 Current Account Deficit and Surplus

A **current account deficit** means the value of imports of goods, services, and income outflows exceeds exports and income inflows. A **surplus** is the reverse.

Causes of a Current Account Deficit

- **Strong domestic demand** sucking in imports (high consumer spending, investment)
- **Overvalued exchange rate** making exports expensive and imports cheap
- **Low productivity/competitiveness** making domestic goods unattractive internationally
- **High relative inflation** reducing price competitiveness
- **Structural factors:** Decline of manufacturing, dependence on imported energy/goods

Significance of a Current Account Deficit

Why it may be a concern:

- Must be financed by capital/financial inflows (selling assets, borrowing from abroad)
- May indicate declining competitiveness
- Persistent deficits lead to rising external debt
- Sudden reversal of capital flows could cause a currency crisis

Why it may not be a concern:

- If financing FDI (productive investment), it builds future export capacity
- Reflects strong economic growth and consumer confidence
- Floating exchange rate may self-correct (deficit depreciates currency, boosting exports)
- The US has run persistent deficits for decades without crisis

Case Study: UK Current Account Deficit

The UK has run a persistent current account deficit since the early 1980s, reaching around 3-5% of GDP in recent years. The UK runs a deficit on trade in goods (importing more manufactured goods and energy than it exports) but a surplus on trade in services (financial services, insurance, consulting — the City of London is a major net exporter of financial services). The deficit is financed by financial inflows — foreign investment in UK property, government bonds, and businesses. The deficit partly reflects the UK's deindustrialisation and shift to a service-based economy. It is sustained because international investors remain willing to hold sterling assets, but Brexit uncertainty has at times weakened this confidence.

6. Globalisation and Trade

Globalisation is the increasing integration and interdependence of the world's economies through increased trade, capital flows, migration, and the spread of technology and ideas.

6.1 Causes of Globalisation

- **Trade liberalisation:** Reduction of tariffs and quotas through the WTO and free trade agreements (e.g. EU Single Market)
- **Technological advances:** Internet, containerisation, cheaper air freight, telecommunications — reducing transport and communication costs
- **Financial deregulation:** Removal of capital controls, enabling free movement of money across borders
- **Multinational corporations (MNCs):** Firms operating across borders, seeking lower costs, new markets, and resources

- **Political factors:** Fall of communism, China joining WTO (2001), EU enlargement

6.2 Benefits of Free Trade

- **Comparative advantage:** Countries specialise in goods they can produce at lowest opportunity cost, increasing total output
- **Lower prices** for consumers through greater competition and access to cheaper imports
- **Greater variety** of goods and services available
- **Economies of scale:** Access to larger markets allows firms to grow and reduce unit costs
- **Technology transfer:** FDI brings new technology, management practices, and skills
- **Dynamic efficiency:** International competition drives innovation

6.3 Costs and Concerns

- **Structural unemployment:** Industries unable to compete with imports may decline (e.g. UK steel, textiles)
- **Inequality:** Gains from trade may be unevenly distributed — skilled workers and capital owners benefit more
- **Environmental damage:** Race to the bottom as countries lower environmental standards to attract investment
- **Loss of sovereignty:** Trade agreements may constrain government policy-making
- **Dependency:** Over-reliance on imports for essential goods (e.g. medical supplies, as exposed by COVID-19)
- **Cultural homogenisation:** Global brands may displace local culture and businesses

Case Study: China's Integration into the Global Economy

China joined the WTO in 2001 and rapidly became the world's largest exporter by 2009. Its integration illustrates both benefits and costs of globalisation. Benefits: hundreds of millions lifted out of poverty (extreme poverty fell from 88% in 1981 to under 1% by 2015), massive investment in infrastructure, technology transfer, and economic growth averaging nearly 10% per year for three decades. Costs: environmental degradation (China became the world's largest CO2 emitter), widening inequality between coastal and inland regions, displacement of manufacturing jobs in

developed countries (the "China shock" is estimated to have cost 2-2.4 million US manufacturing jobs between 1999 and 2011), and concerns over intellectual property practices.

6.4 Comparative Advantage

Comparative advantage (Ricardo): A country has a comparative advantage in producing a good if it can produce it at a lower opportunity cost than another country. Even if one country is more efficient at producing everything (absolute advantage), both countries benefit from specialisation and trade based on comparative advantage.

Example: If Country A can produce either 100 cars or 200 tonnes of wheat, and Country B can produce either 40 cars or 120 tonnes of wheat, then:

- Country A's opportunity cost of 1 car = 2 tonnes of wheat
- Country B's opportunity cost of 1 car = 3 tonnes of wheat
- Country A has a comparative advantage in cars (lower opportunity cost); Country B in wheat
- Both benefit if A specialises in cars and B in wheat, then they trade

Limitations of Comparative Advantage Theory

- Assumes constant costs of production (ignoring economies of scale)
- Assumes perfect factor mobility within countries and immobility between them
- Ignores transport costs and trade barriers
- Assumes complete specialisation — unrealistic in practice
- Static model — does not account for changing comparative advantages over time
- Assumes homogeneous goods and perfect competition

Exam tip: When discussing macroeconomic performance, always consider the **trade-offs** between objectives. For example, policies to reduce inflation may increase unemployment; policies to promote growth may worsen the current account deficit. These conflicts are a key evaluation theme.

Evaluation — Macroeconomic Objectives and Trade-offs

- The four main macroeconomic objectives are: economic growth, low unemployment, low and stable inflation, and a sustainable balance of payments.
- These objectives often conflict: rapid growth may cause inflation; reducing inflation may increase unemployment; growth may worsen the current account.
- Policy-makers must prioritise — which objective matters most depends on the current economic context.
- There is no consensus on the "right" policy mix — Keynesian and classical economists disagree fundamentally on the role of government intervention.

7. The AD/AS Model

Aggregate Demand (AD): The total planned expenditure on goods and services in an economy at a given price level: AD = C + I + G + (X − M). The AD curve slopes downward — a lower price level increases real wealth, reduces interest rates (via money demand), and improves international competitiveness, all boosting spending.

Short-Run Aggregate Supply (SRAS): The total output all firms are willing and able to produce at each price level in the short run. Slopes upward because wages and input costs are sticky — higher prices improve profit margins and incentivise more production.

Long-Run Aggregate Supply (LRAS): The economy's productive potential — the maximum sustainable output when all resources are fully and efficiently employed. Classical economists argue it is vertical at the full-employment level of output (Yfe).

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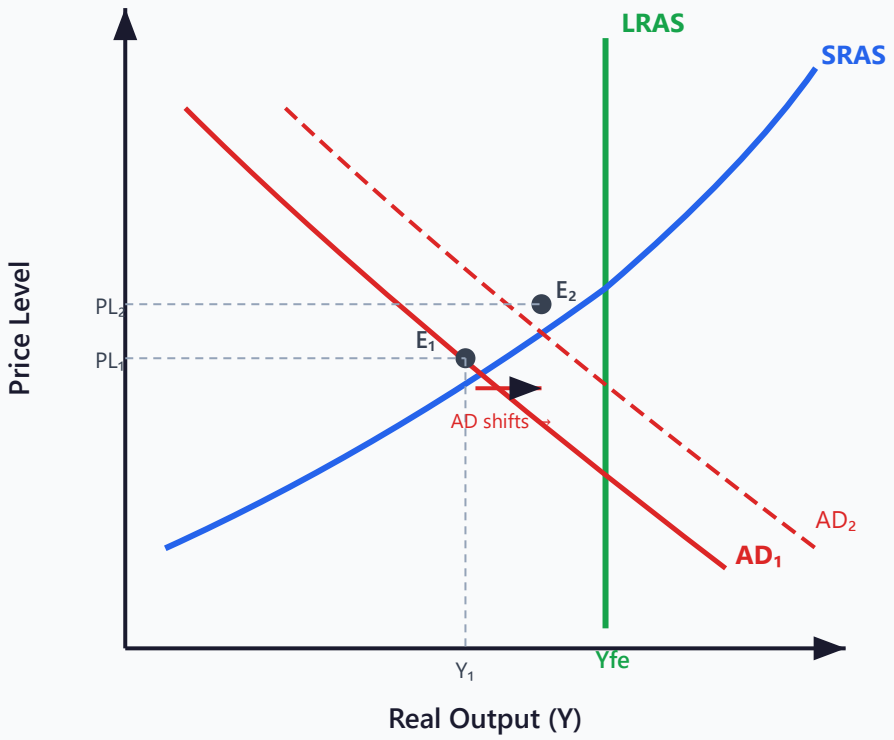

Figure 2: The AD/AS model. LRAS is vertical at Yfe (full employment). An increase in AD from AD_1 to AD_2 raises both the price level (PL_1 to PL_2) and real output (Y_1 closer to Yfe) in the short run. If the economy is already at Yfe, the AD increase causes only inflation.

7.1 Shifts of AD and SRAS — Key Macro Scenarios

Scenario	What Shifts	Effect on Price Level	Effect on Output	Real-World Example
Demand-pull inflation	AD right (near Yfe)	Rises	Little increase (at capacity)	UK 2021 post-lockdown boom
Recession	AD left	Falls (or disinflation)	Falls — unemployment rises	UK 2008-09; COVID 2020

Cost-push inflation / Stagflation	SRAS left	Rises	Falls simultaneously	UK 2022 energy shock; 1970s oil crisis
Long-run growth	LRAS right	Stable or falls	Rises (productive capacity)	China 1990-2020; East Asian Tigers

Evaluation — AD/AS Model: Keynesian vs Classical View

- **Classical view (vertical LRAS):** The economy is self-correcting — if AD falls, wages and prices fall, restoring full employment automatically. Government intervention is unnecessary and potentially harmful (crowding out). The LRAS determines long-run output; AD only affects the price level.
- **Keynesian view:** Wages are "sticky downward" — workers resist nominal wage cuts. The economy can get stuck in a recession with persistent unemployment and spare capacity. In this zone (left of Yfe), the LRAS is effectively horizontal and increases in AD raise output without inflation. Government intervention through fiscal policy is essential to restore full employment.
- **Practical consensus:** Most economists accept that in the short run, AD fluctuations affect real output (Keynesian insight) and in the long run, productive capacity (LRAS) determines output (Classical insight). The policy challenge is managing short-run fluctuations while supporting long-run LRAS growth.
- **Supply-side shocks complicate the model:** Cost-push shocks create stagflation — a simultaneous fall in output and rise in prices. This is the hardest scenario for policy because raising interest rates reduces inflation but worsens the output fall, while cutting rates helps output but worsens inflation. The 2022-23 episode showed the real difficulty of using monetary policy against supply-side inflation.

8. Protectionism and Free Trade Arguments

Argument for Free Trade	Argument for Protectionism

Comparative advantage — specialisation increases global output	Infant industry protection — new industries need time to achieve economies of scale
Consumer welfare — lower prices, more choice	Job protection — prevents structural unemployment from import competition
Dynamic efficiency — competition drives innovation	National security — strategic industries (defence, food, energy) must be maintained domestically
Economies of scale — larger markets allow lower costs	Anti-dumping — prevents foreign firms pricing below cost to destroy domestic competitors
Technology transfer through FDI and trade	Revenue — tariffs are an important revenue source for developing countries

Evaluation — Free Trade vs Protectionism

- Free trade increases total global welfare in theory, but the distribution of gains matters enormously — workers in import-competing industries bear concentrated losses while consumers enjoy dispersed gains. This political economy dynamic explains protectionist pressures even when free trade is economically optimal overall.
- The infant industry argument is theoretically valid but practically weak — protected industries often fail to become competitive, and protection tends to be permanent rather than temporary (political economy of lobbying).
- The US-China trade war (2018 onwards, 25% tariffs on hundreds of billions of goods) imposed costs of approximately $831 per US household per year (Federal Reserve NY estimate) while failing to significantly reduce the bilateral trade deficit. This illustrates that tariffs primarily redistribute income (from consumers to protected producers and the government) rather than achieving macroeconomic goals.
- Brexit's introduction of non-tariff barriers to EU trade despite a zero-tariff deal illustrates that administrative barriers can be as damaging as tariffs — UK goods trade with EU fell approximately 7% relative to a counterfactual without Brexit (Centre for European Reform, 2023).
- Strategic trade theory (Krugman, Brander-Spencer) shows that in industries with increasing returns to scale, government intervention can shift oligopoly profits to domestic firms — the

basis for industrial policy in tech and aerospace (Airbus vs Boeing).

Exam Tips — OCR National and International Economy

- **GDP limitations:** When asked to evaluate GDP as a welfare measure, structure your answer: inequality (distribution), informal economy (shadow), non-market production (household work), externalities (pollution), composition (military vs healthcare). Use HDI as a superior alternative and explain why.
- **Inflation questions:** Always distinguish demand-pull (AD rightward shift) from cost-push (SRAS leftward shift) — they have different policy implications. Cost-push is harder to address because the standard tool (raising interest rates) worsens the output loss without eliminating the supply-side cause.
- **Balance of payments:** A current account deficit is not automatically bad — assess how it is financed (FDI is more sustainable than short-term capital flows), what caused it (strong growth vs low competitiveness), and whether it is self-correcting (floating exchange rate can depreciate to restore competitiveness).
- **Comparative advantage:** Use a numerical example in your answer — calculate opportunity costs to demonstrate the advantage. Always note the key assumptions (constant costs, no transport costs, full factor mobility, no externalities) and why they limit the theory's real-world applicability.

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